AI531C — T13 · The Metrics Clinic: Student Sheet

Tue 10 Nov, 2:00–3:00 PM · run: python3 T13_studio.py (+ 'support' / 'extension')

Your mission

Execute the full segmentation-metric protocol on a demo pair, then on YOUR masks — floors first, boundary band last, conventions stated.

Experiment

Demo pair (coins GT vs 8×-coarse prediction): acc 0.955, Dice 0.945, IoU 0.896, sens 0.939 — then the boundary band: 0.694. The gap IS the story (L36). Then: your masks, your three counts, same table.

Vary / break it

Vary the band width 3/7/15 px on the demo — how sensitive is the boundary metric to the convention? (State the width you report!)

Discuss

Q1 · All-foreground scored Dice 0.580 on this dataset 'for free'. What dataset property sets that floor?

Q2 · Overall IoU 0.896 vs band 0.694 — write the one-sentence diagnosis and the architectural cause.

Q3 · Two papers report 'boundary IoU' with different band widths. Are their numbers comparable? What must the methods section state?

Reflect

Whatever your lane, this table format is your results section — frozen from today.

Fast? EXTENSION cell: per-object Dice on the demo (worst 0.94, mean 0.96) — which coins fail and why? Struggling? SUPPORT cell: toy numbers |A|=150 |B|=120 overlap 90 → Dice 0.667, IoU 0.5, identity check. Nothing is due after class — the hour is the work.